

engineering solutions to various issues related to ports, waterways, and coasts in the country, government has set up the **National Technology Centre for Ports, Waterways and Coasts (NTCPWC)** at IIT Madras apart from establishing the Centre for Inland and Coastal Maritime Technology (CICMT) at IIT Kharagpur which serves as the government's technology arm to provide research, testing and experimentation facility to IWAI, CSL and major ports.

A living testimony to Hon'ble PM's vision of self reliant India, these indigenous state of the art institutes are already **competing with World class players in providing cost effective consultancy** and technical services to ports across the country building next generation technology like night navigation at ports, etc.

COASTAL SHIPPING AND INLAND WATERWAYS

Cargo movement through coastal shipping and inland waterways is sustainable and environment friendly. Hence, an impetus has been given by the government under the Sagarmala Programme to undertake projects which will result in shift of cargo from land-based transportation to coastal and inland waterways mode.

The sector posed many key challenges such as non-availability of return cargo, lack of dedicated coastal shipping infrastructure, dependency of first and last mile connectivity on other modes, insufficient parcel size for transporting through vessel, etc. In this respect, government has launched several projects to address these challenges, resulting in a significant increase in the coastal cargo volume handled from **74.9 to 133 million tonnes**. Further to boost the coastal infrastructure under Sagarmala Programme, government has completed constructed **5 coastal berths worth Rs 122 crore,** adding a net capacity of 6.35 million tonnes per annum.

A major boost to **Inland Waterways Transport** sector has been provided by the Government through enactment of National Waterways Act, 2016 taking the total number of National Waterways (NW) from **5 in 2014 to 111 in 2022** which now cover a total length of **more than 20,000 kms** spreading across 24 States in the country. Owing to various Infrastructural, regulatory and technology-led initiatives by government to promote Inland Waterways, the sector witnessed meteoric rise in cargo volume handled from ~**55 million tonnes** in 2016-17 to about **109 million tonnes** in 2021-22.

Government has been actively working towards increasing India's share of Intra-BIMSTEC cargo trade via Inland Waterways by promoting cargo movement to Bangladesh, Bhutan and Nepal. Through many government led initiatives, cargo Services via Inland Waterways from Haldia to Patna and Dhubri (Assam) to Bangladesh have now started. In terms of demand activation, several oil and petro-chemical companies have already pledged long-term cargo movement for the first time from Assam to Bangladesh and Haldia via Inland Waterways. Moreover, the upgradation of Badarpur and Karimganj terminals on river Barak (NW-16) in Lower Assam has boosted the cargo movement between India and Bangladesh.

A pilot movement of 200 ton food grains belonging to Food Corporation of India (FCI) was successfully undertaken from Patna (Bihar) to Pandu (Guwahati), Assam via Indo-Bangladesh Protocol route covering a distance of 2,350 km.

As part of regularised movement of stone aggregates from Bhutan through Dhubri (Assam) IWT terminal, 319 vessels loaded with stone aggregates / coal of total weight 51,210 MT sailed from Dhubri (India) to Chilmari (Bangladesh) during FY 2021-22. Infra and business improvement resulted in 5.43 million tonnes of cargo movement during FY21-22 on Indo Bangladesh Protocol route which was all time high.

Further, Inland Waterways Authority of India (IWAI), under Ministry of Ports, Shipping & Waterways (MoPSW), has identified 25 viable NWs for further development and increasing the cargo handled by inland waterways. Moreover, the government under Sagarmala Programme is implementing **37 Inland Waterways projects** of

Hon'ble Union Minister launching Battery Electric Autonomous Vessel delivered to Norway by CSL





> "Under Sagarmala Programme, we have a huge pipeline of infrastructure projects for holistic development of entire coastline. Also, focussed efforts are being made for enhancing the role of Coastal Shipping & Inland Waterways in the country. Further, we have identified more than 200 port connectivity projects for providing seamless port connectivity to the hinterland. We are preparing a blueprint to expand our port capacities by 4X to 10,000 million tonnes to cater to the rising cargo growth."
>
> – Bhushan Kumar, Joint Secretary, Sagarmala & PPP, Ministry of Ports, Shipping and Waterways



Port Rail Connectivity



Newly Constructed Floating Jetty